MILESTONE 3 · DATA SCIENCE AND AI LAB · IIT MADRAS BS

Socratic AI Tutor for NCERT Classes 6–10

Model selection, baselines, hyperparameter tuning and the end-to-end pipeline for the tutor's three learned components: the Socratic response model, knowledge tracing and textbook retrieval.

Qwen 3.5-4B

Selected Socratic model · ROUGE-L 0.19 → 0.33, 0 answer leaks

DKT

Knowledge tracing · 7 architectures compared on ASSISTments

124 passages

NCERT retrieval · cited, built automatically from 4 chapters

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One tutor, three learned components

The tutor guides a learner to the answer with questions instead of telling it. Each part of that behaviour is a separate modelling problem, so we compared architectures for each one on a small, representative candidate set before any full-scale training.

SOCRATIC RESPONSE MODEL

What should the tutor say next?

Decoder-only LLMs (≤4B) with QLoRA adapters, trained on tutoring dialogues.

Compared: Qwen 3.5-4B, Gemma 4 E4B, Phi-4-mini, Qwen3-4B-Instruct (trial)

Selected: Qwen 3.5-4B, LoRA r16, lr 2e-4

KNOWLEDGE TRACING

What does the learner already know?

Sequence models that predict the next answer's correctness per skill.

Compared: DKT (LSTM, GRU, RNN), SAKT, SAKT + positions, AKT, AKT + Rasch

Selected: DKT (LSTM), AUC 0.829; AKT leads (0.843) but needs question IDs

TEXTBOOK RETRIEVAL (RAG)

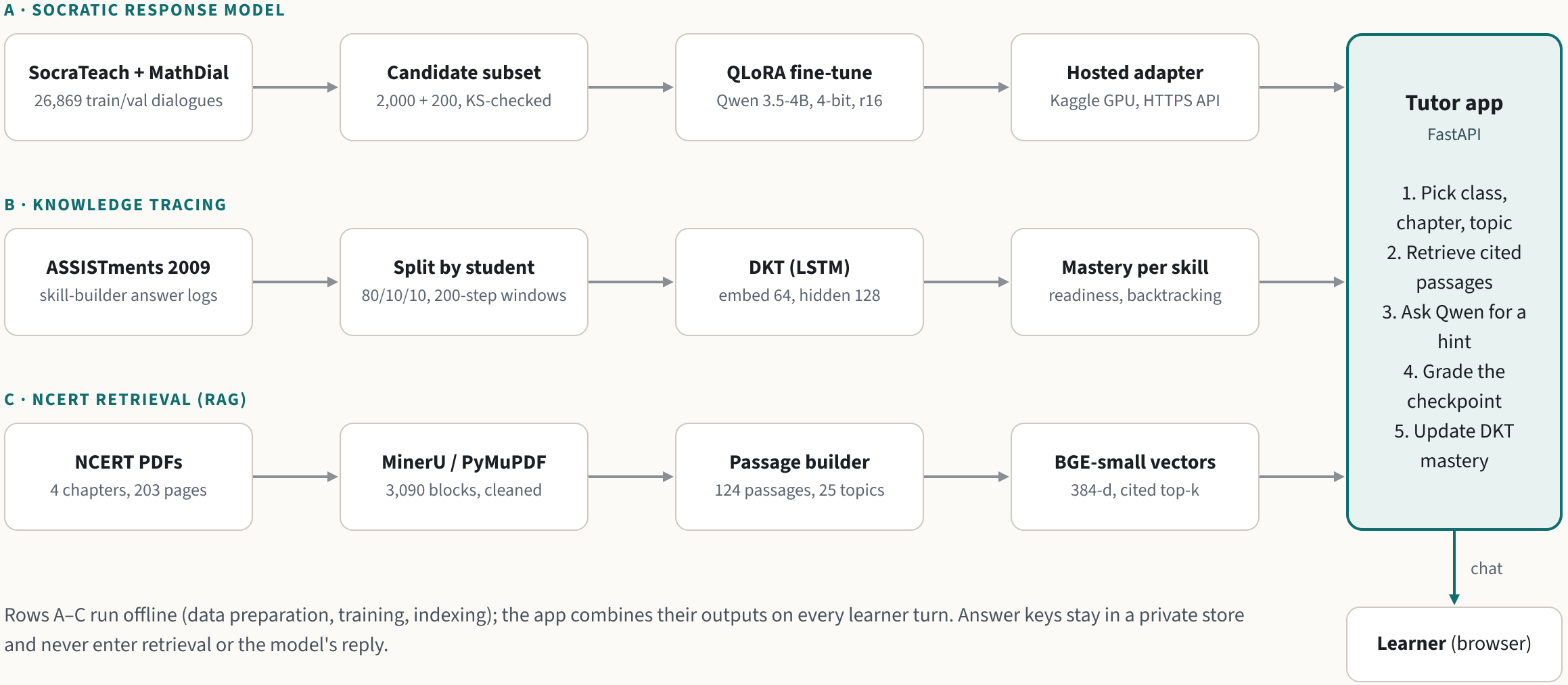
What does NCERT actually say?

Parsed NCERT chapters, embedded passages, cited top-k retrieval.

Compared: PyMuPDF vs MinerU parsing; BM25, BGE-small dense, BM25 + BGE fusion

Selected: MinerU for maths, BM25 + BGE fusion

End-to-end pipeline: data → models → tutor app



Rows A–C run offline (data preparation, training, indexing); the app combines their outputs on every learner turn. Answer keys stay in a private store and never enter retrieval or the model's reply.

2,000 + 200 dialogues that match the full split, source by source

How it was built

1. Raw SocraTeach_Multi (35,151 dialogues) and MathDial (2,262 + 599) at pinned revisions, unified into one schema with problem, answer and per-turn leak flags.
2. Split by **problem**: all 599 MathDial official-test dialogues held out; every dialogue sharing a test problem removed; 10% of problems to validation; ≤3 dialogues per problem.
3. Candidate subset drawn from train and val, **stratified by word-count quartile** within each source, seed 42; membership asserted against the parent split.

Split	SocraTeach	MathDial	Use
Full train	22,574	1,302	Final training (M4)
Full val	2,682	311	Final validation
Test	—	599	Untouched, final evaluation
candidate_train	**1,000**	**1,000**	Grid search + final LoRA
candidate_val	**100**	**100**	100 eval contexts, val loss

Why it is representative: two-sample KS test vs full train

Source · feature	Full	Subset	KS	p
SocraTeach · words / dialogue	234.9	234.3	0.020	0.81
SocraTeach · messages	9.71	9.69	0.013	0.99
MathDial · words / dialogue	314.2	312.9	0.010	1.00
MathDial · messages	12.04	12.03	0.013	1.00
MathDial · leaking tutor turns	0.017	0.017	0.004	1.00
MathDial · "telling" turns	0.154	0.153	0.006	1.00

Deliberate source mix. MathDial is 5.5% of full train but 50% of the subset: it is the only real-teacher data and the whole test set. At its natural share the subset would hold ~110 MathDial dialogues, too few to separate configurations. Results are therefore always reported per source.

Source: socratic_data_prep/REPORT.md, processed/stats.json

Small open LLMs + QLoRA, compared under one protocol

Candidate	Size	Architecture notes	LoRA targets	Licence
Qwen 3.5-4B (selected)	4B	Hybrid linear + full attention; thinking mode off	q, k, v, o, gate, up, down	Apache 2.0
Gemma 4 E4B	~4B effective	Multimodal base; vision/audio towers excluded from LoRA	language-model q…down	Gemma terms
Phi-4-mini-instruct	3.8B	Phi-3 family, fused qkv and gate_up projections	qkv, o, gate_up, down	MIT
Qwen3-4B-Instruct-2507	4B	Early feasibility trial, different split; loss only, not ranked	q…down (α 32)	Apache 2.0

Why this approach

- The gap is **behaviour**: untrained models explain at 2–2.5× tutor length and some state the answer.
- Fine-tuned small models told students the answer far less than prompted ChatGPT in MathDial (Telling@5 <1% vs 16%) [3].
- Fits one free 16 GB T4; student data stays on team-hosted infrastructure.

Design decisions

- Loss on tutor turns only; leaking and "telling" turns kept as context, never as targets.
- Correct answer in the system prompt *for checking only*; leak rate measured.
- End-of-turn token trained so the model stops after one turn.
- QLoRA: 4-bit NF4 base, 10–35M trainable parameters [1, 2].

Trade-offs

- 7–8B models would likely score higher but don't train on a T4 at this length.
- 768-token cap cuts ~22% of dialogues back to the last tutor turn that fits.
- T4 has no bf16: fp16 with a finite-loss check before training.
- ROUGE/BLEU reward wording, so leak, question rate and length are reported beside them.

[1] Hu et al. 2022, LoRA · [2] Dettmers et al. 2023, QLoRA · [3] Macina et al. 2023, MathDial

Untrained models explain and solve instead of asking

Pretrained weights in 4-bit, same 100 validation contexts (50 per source), greedy decoding, ≤96 new tokens.

Baseline	Leak	Question	Words	ROUGE-L	BLEU	MathDial ROUGE-L
Real tutors	0.01	0.62	22.2	—	—	—
Gemma 4 E4B	0.00	0.87	48.0	0.189	0.040	0.111
Qwen 3.5-4B	0.01	0.52	56.4	0.190	0.040	0.124
Phi-4-mini	0.06	0.60	39.6	0.226	0.071	0.137

- Replies are **2–2.5× longer** than a real tutor's: the models explain rather than probe.
- Phi-4-mini leaks the answer in 6% of replies overall and **12% on MathDial**.
- MathDial (student has made an error) is hardest: ROUGE-L 0.11–0.14.

Metrics. Leak rate: reply states the final numeric answer before the student has. Question rate: reply contains a question. Words: reply length. ROUGE-L [6] and BLEU [7] (sacreBLEU ÷ 100): overlap with the real tutor's next turn. Validation loss: compared only within a model (tokenizers differ).

Industry benchmarks (reference only)

Source	Setting	Result
MathDial [3]	Flan-T5 780M fine-tuned, MathDial test	sBLEU 9.7; Telling@5 <1% vs ChatGPT 16%
SocraticLM [4]	ChatGLM3-6B on SocraTeach, vs GPT-4	ROUGE-L 47.5 vs 32.4; BLEU-4 48.6 vs 36.2
Ours, tuned	Qwen 3.5-4B, 50 SocraTeach contexts	ROUGE-L 41.9; BLEU 21.7
Ours, tuned	Qwen 3.5-4B, 50 MathDial contexts	ROUGE-L 24.4; sBLEU 5.6

Splits and metrics differ from the papers, so these frame our numbers rather than rank them. Our tuned SocraTeach ROUGE-L sits between GPT-4 and SocraticLM's fine-tuned 6B model. A like-for-like MathDial number comes from the untouched 599-dialogue test split in M4.

[3] Macina et al. 2023 · [4] Liu et al. NeurIPS 2024 · [6] Lin 2004 · [7] Papineni et al. 2002; Post 2018

Grid search on 400 dialogues: rank 16 at 2e-4 wins for all three models

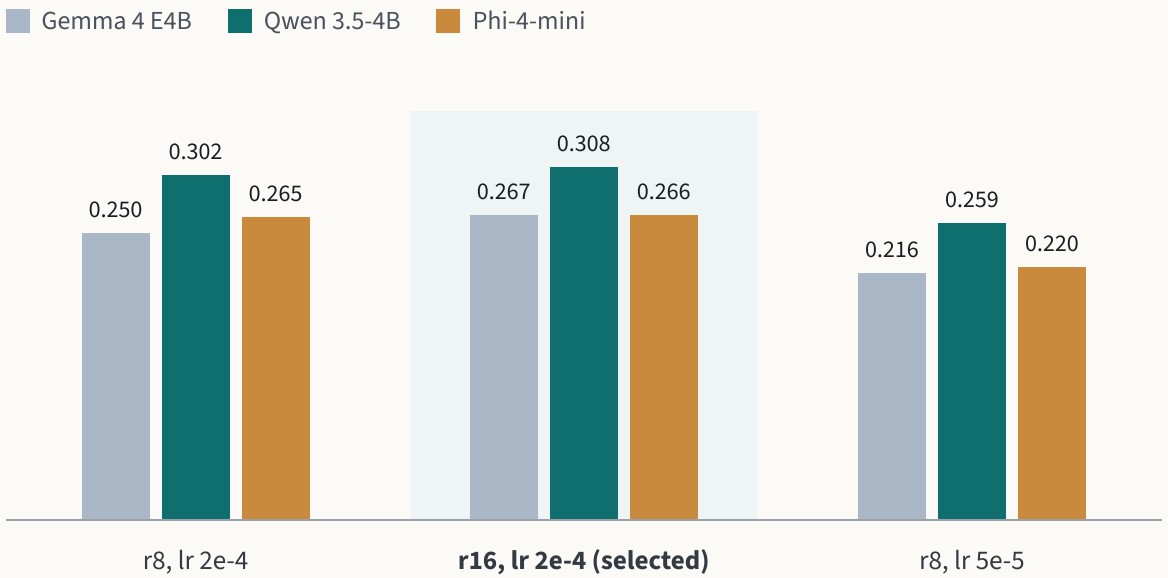
Search strategy: grid search, 3 settings × 3 models. Each setting trains 1 epoch on the **same 400 candidate dialogues** (~50 optimizer steps) and is scored on the same 100 contexts.

Hyperparameter	Tried	Rationale
LoRA rank r (α = r)	8, 16	Does more adapter capacity help? α = r keeps scale 1
Learning rate	2e-4, 5e-5	2e-4 is the QLoRA default [2]; 5e-5 tests a gentler update
Fixed	dropout 0.05 · cosine, 5% warm-up · paged AdamW 8-bit · effective batch 8 · max 768 tokens · 1 epoch · seed 42	

Selection rule, fixed before training: lowest validation loss among settings whose leak rate is at most 2 points above real tutors (0.01 + 0.02).

- **Low learning rate under-trains:** highest val loss and lowest ROUGE-L for every model.
- **Rank 16 beats rank 8** at 2e-4 on val loss for all three, at the same 10–19 min training cost.

ROUGE-L by setting (400-dialogue runs)

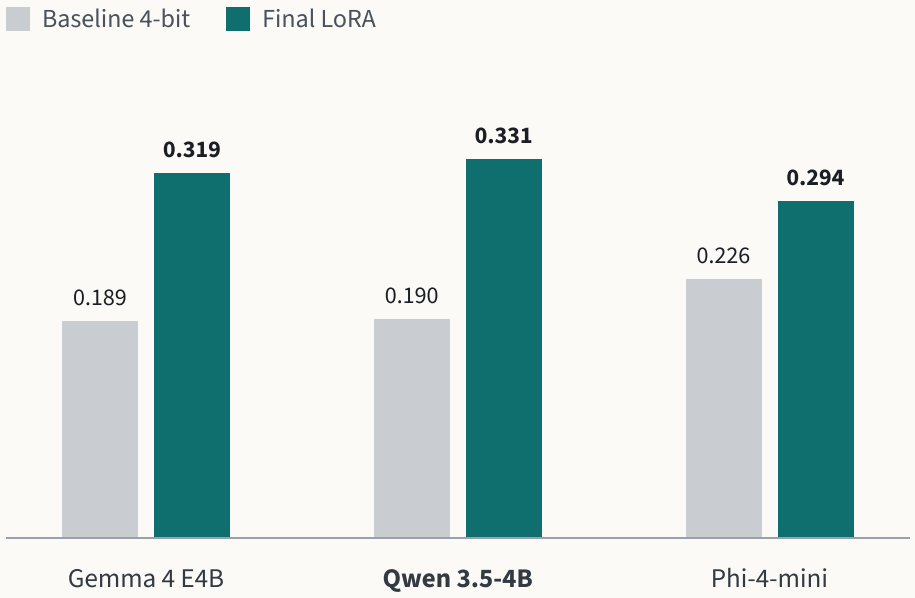


Val loss / leak	r8, 2e-4	r16, 2e-4	r8, 5e-5
Gemma 4 E4B	1.260 / 0.02	**1.233 / 0.00**	1.406 / 0.03
Qwen 3.5-4B	1.169 / 0.00	**1.154 / 0.00**	1.255 / 0.01
Phi-4-mini	1.273 / 0.01	**1.256 / 0.01**	1.403 / 0.00

Source: results.json in each *_finetune_output folder · [2] Dettmers et al. 2023

Fine-tuning removes leaks, halves reply length and lifts overlap 30–74%

ROUGE-L, baseline → final LoRA (r16, 2e-4, ~2,000 dialogues)



Selected: Qwen 3.5-4B. Best overall and on MathDial, no leaks, question rate per source closest to humans, Apache 2.0, already served to the app. With 100 contexts the three are a near tie; the MathDial test split confirms in M4.

Final model	Leak	Question (0.62)	Words (22.2)	ROUGE-L	BLEU	Val loss	Train
Gemma 4 E4B	0.00	0.69	20.5	0.319	0.175	2.76 → 1.12	62 min
Qwen 3.5-4B	**0.00**	0.59	20.3	**0.331**	**0.176**	1.82 → 1.07	80 min
Phi-4-mini	0.00	0.65	21.8	0.294	0.174	1.85 → 1.17	**49 min**

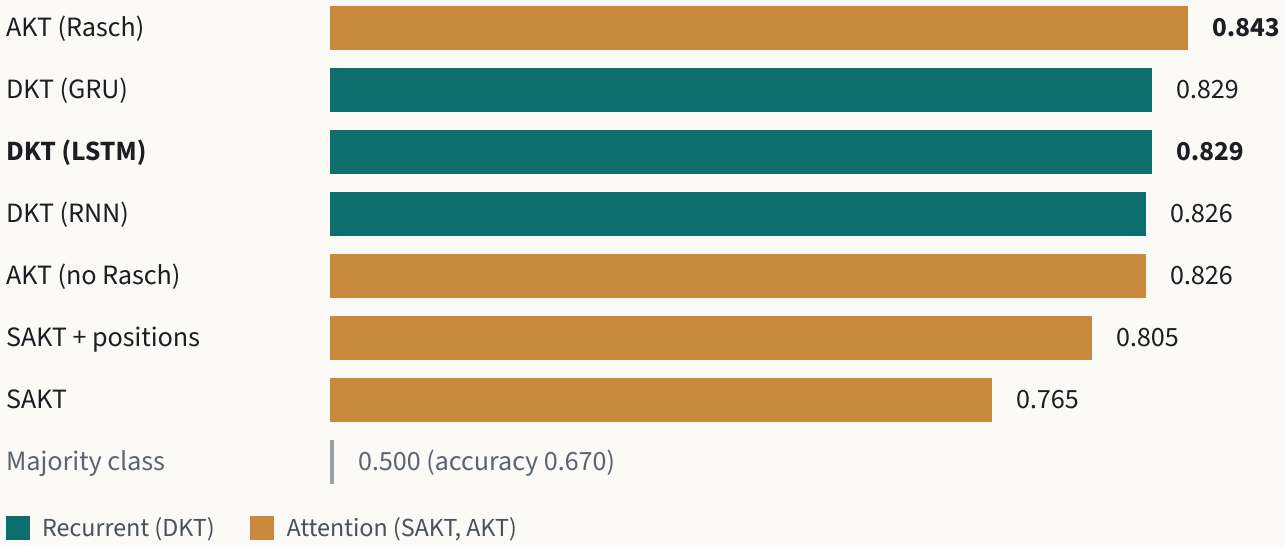
Per source (50 each)	SocraTeach ROUGE-L	MathDial ROUGE-L	SocraTeach Q (0.80)	MathDial Q (0.44)
Gemma 4 E4B	0.409	0.228	0.80	0.58
Qwen 3.5-4B	**0.419**	**0.244**	0.76	**0.42**
Phi-4-mini	0.402	0.186	0.80	0.50

- All three reach **human-like length** (20–22 words, from 40–56) and **zero leaks** (Phi from 6%).
- Qwen gains most (+74% ROUGE-L); Phi starts strongest but gains least (+30%) and trails on MathDial.
- Val loss is compared within each model only; no empty or runaway replies in any run.

Same 100 validation contexts for every model · values in parentheses are real tutors

AKT leads by 0.015 AUC; DKT is kept because NCERT has no question IDs

Test AUC on ASSISTments 2009 (axis from 0.50 = chance)



27,329 test predictions from unseen students. One seed: gaps of a few thousandths are ties.

Data. ASSISTments 2009 skill builder: 123 skills, 15,922 questions; split by student 80/10/10; 200-step windows.

Tuning. Shared defaults for all seven: Adam 1e-3, batch 64, clip 5, up to 60 epochs, early stop after 8 without a val-AUC gain, best epoch restored.

Model	Params	Best ep.	Val AUC	F1
DKT (LSTM)	131K	11	0.826	0.846
AKT (Rasch)	491K	16	0.836	0.848
SAKT + positions	62K	27	0.801	0.842

Why DKT. AKT's lead comes from per-question Rasch difficulty, which needs ASSISTments question IDs. Without it AKT scores 0.826, below DKT (0.829). DKT has about a quarter of the parameters and updates once per answer.

Benchmark: DKT first reported ~0.86 on ASSIST09 [8], about 0.75 after duplicate rows were removed [11]. We keep multi-skill rows, so our absolute AUCs are likely optimistic; the ranking is fair.

An automatic textbook pipeline; BM25 + BGE fusion puts the right passage in the top 5 for 80% of test questions

1 · Parse
NCERT PDFs → MinerU or PyMuPDF, routed per page

→

2 · Clean
drop page numbers, running heads, broken markup

→

3 · Passages
merge by numbered section, 300-token target, 400 max

→

4 · Questions
chapter questions matched to answer-page keys; figure items skipped

→

5 · Index
BGE-small 384-d in pgvector + BM25; frozen, fingerprinted release

Release ncert-pilot-a2ac2147	Count
Passages (from 792 cleaned blocks)	124
Topics the app can open	25
Questions approved / held	84 / 1
Gradable question–key pairs	43
Figure-dependent questions left out	16
Automated tests passing	52

Design decisions. No manual chunking: the same builder runs for Classes 6–10. Answer keys stay private to the grader and never enter the prompt. Figure questions wait until the app can show the figure.

Retrieval on 15 held-out textbook queries

Retriever	Hit@1	Hit@5	MRR
BM25 (keywords)	**53.3%**	73.3%	**0.597**
BGE-small (dense) [12]	46.7%	73.3%	0.572
Rank fusion (used)	46.7%	**80.0%**	0.591

- Fusion has the best Hit@5. Labels cover only the 6 original concepts, so some "misses" land on correct new topics; labels were not changed after seeing results.
- **Automatic topic routing is off** (0/15 accepted at the calibrated cut-offs): the learner picks the topic and the app ranks passages inside it (top 2 by BGE cosine).
- Misconception retrieval on 300 Eedi groups: Recall@5 **16% → 26%** with a contrast query + BM25; too weak to label learners, so suggestions stay unconfirmed.
- Limit: 15 queries is small; a test labelled against all 25 topics is the M4 check.

Source: Ncert work/PILOT_REPORT.md · [12] Xiao et al. 2023, BGE embeddings

One learner turn: topic → passages → Qwen → screen → grade → mastery

1 · Choose
Class → Chapter → Topic in the UI

→

2 · Route intent
explain, hint (3 levels), discuss or check answer

→

3 · Retrieve
top 2 NCERT passages in the topic, BGE cosine

→

4 · Generate
Qwen 3.5-4B + LoRA, hosted behind an HTTPS tunnel

→

5 · Screen
block answer leaks, bad protocol, truncation

→

6 · Grade + trace
private key check, DKT readiness, prerequisite backtrack

Stage	Socratic model	Knowledge tracing	NCERT retrieval
Input data	SocraTeach + MathDial → chat format, loss on tutor turns, 768 tokens	ASSISTments 2009 → per-student sequences, 200 steps	9 pinned NCERT PDFs → passages, questions, private keys
Training	QLoRA on Kaggle T4; 3-setting grid on 400 dialogues → final r16 on ~2,000	7 architectures, shared defaults, up to 60 epochs with early stopping	No training: BGE-small frozen; BM25 index; calibrated cut-offs
Validation	Val loss each run; 100 fixed contexts, 50 per source	Val AUC each epoch; best epoch restored	24 calibration questions; release gates and fingerprints
Evaluation	Leak, question rate, words, ROUGE-L, BLEU vs baseline	Test AUC, accuracy, F1 once at the end	Hit@1, Hit@5, MRR on 15 held-out; Eedi Recall@5
Output	One short Socratic turn (~20 words) that does not state the answer	Per-skill probability of a correct next answer	Cited passages ("book · PDF page") and a gradable checkpoint

Built and checked: FastAPI app with SQLite learner history; unit tests plus `check_offline.py` run on the real release with zero model calls; failed model calls keep progress and can be retried.

Not yet tested: the live Qwen step with retrieved passages (`check_live.py`, M4). DKT predicts only for NCERT topics mapped to an ASSISTments skill; others use practice estimates.

Source: mock-app/README.md, Ncert work/PILOT_REPORT.md

Each part works on its own; M4 tests them together on untouched data

Known limitations

- **Small evaluation:** 100 contexts per model and one seed; Qwen vs Gemma is a near tie.
- **Overlap metrics** reward wording, not teaching; no human or LLM-judge rating yet.
- **Domain gap:** tutoring data is maths word problems in English; NCERT science is not covered by fine-tuning.
- **KT:** one seed, untuned, multi-skill rows kept, no BKT baseline; trained on ASSISTments skills, not NCERT topics.
- **Retrieval test:** 15 held-out questions, labels for 6 of 25 topics; automatic topic routing is off.
- **Leak guard** is lexical; it does not prove a reply is safe.

Plan for Milestone 4

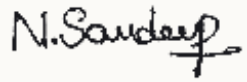
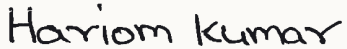
- Score all models once on the **599-dialogue MathDial test split**, with Telling@k as in [3].
- Add a rubric rating (human sample + LLM judge) for guidance quality.
- Run `check_live.py`: Qwen with retrieved NCERT passages, end to end.
- Extend the passage builder to all Classes 6–10 books; label a fresh retrieval test against every topic.
- Collect in-app answers to fit or calibrate KT on NCERT topics.
- Re-test topic routing once the larger labelled set exists.

[3] Macina et al. 2023, MathDial

Group 2 · Data Science and AI Lab, IIT Madras BS

Member	Primary role	Milestone 3 work
Sandeep Nalla 21f1006125	Backend	Zero-shot baseline evaluation script; Phi-4-mini QLoRA grid and final run; Phi adapter on Hugging Face
Hariom Kumar 23f2004225	RAG agent	Team coordination; data prep and candidate set; Gemma 4 E4B and Qwen 3.5-4B fine-tuning and four-model comparison; NCERT pilot; demo app; Qwen hosting; dataset and models on Hugging Face; report and slides
Hariom Pandey 21f3002112	Fine-tuned models	Knowledge-tracing comparison of DKT, SAKT and AKT (with Mrityunjay); Qwen3-4B fine-tuning trial
Mrityunjay Chakraborty 21f3001385	Frontend	Knowledge-tracing comparison of DKT, SAKT and AKT (with Hariom Pandey)

Models and dataset: huggingface.co/IITM-2026T3-DSAI-GROUP2

Sign-off	Roll number	Signature	Date
Sandeep Nalla	21f1006125		05-11-2026
Hariom Kumar	23f2004225		05-11-2026
Hariom Pandey	21f3002112		05-11-2026
Mrityunjay Chakraborty	21f3001385		05-11-2026